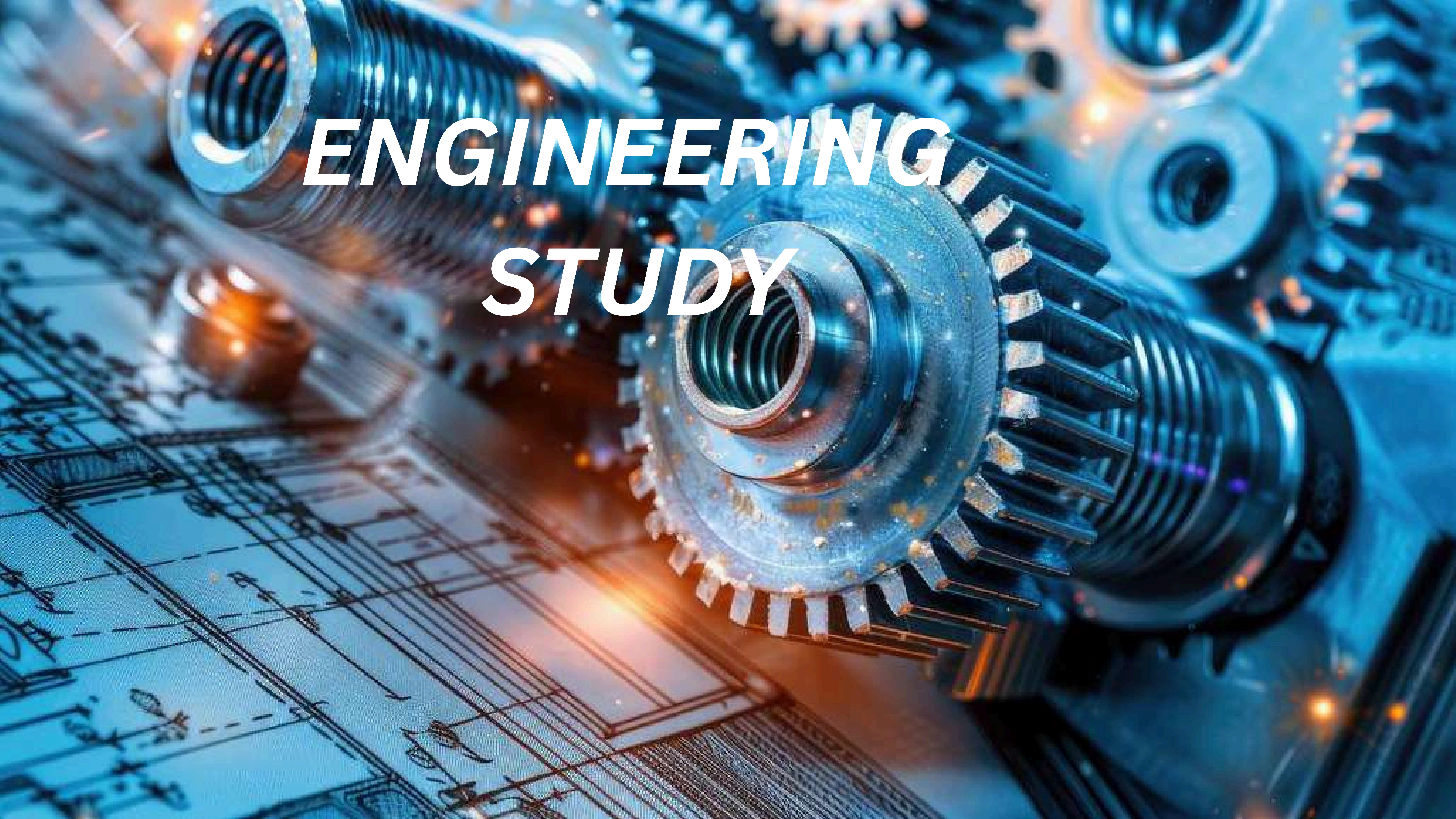
ENGINEERING STUDY



ENGINEERING

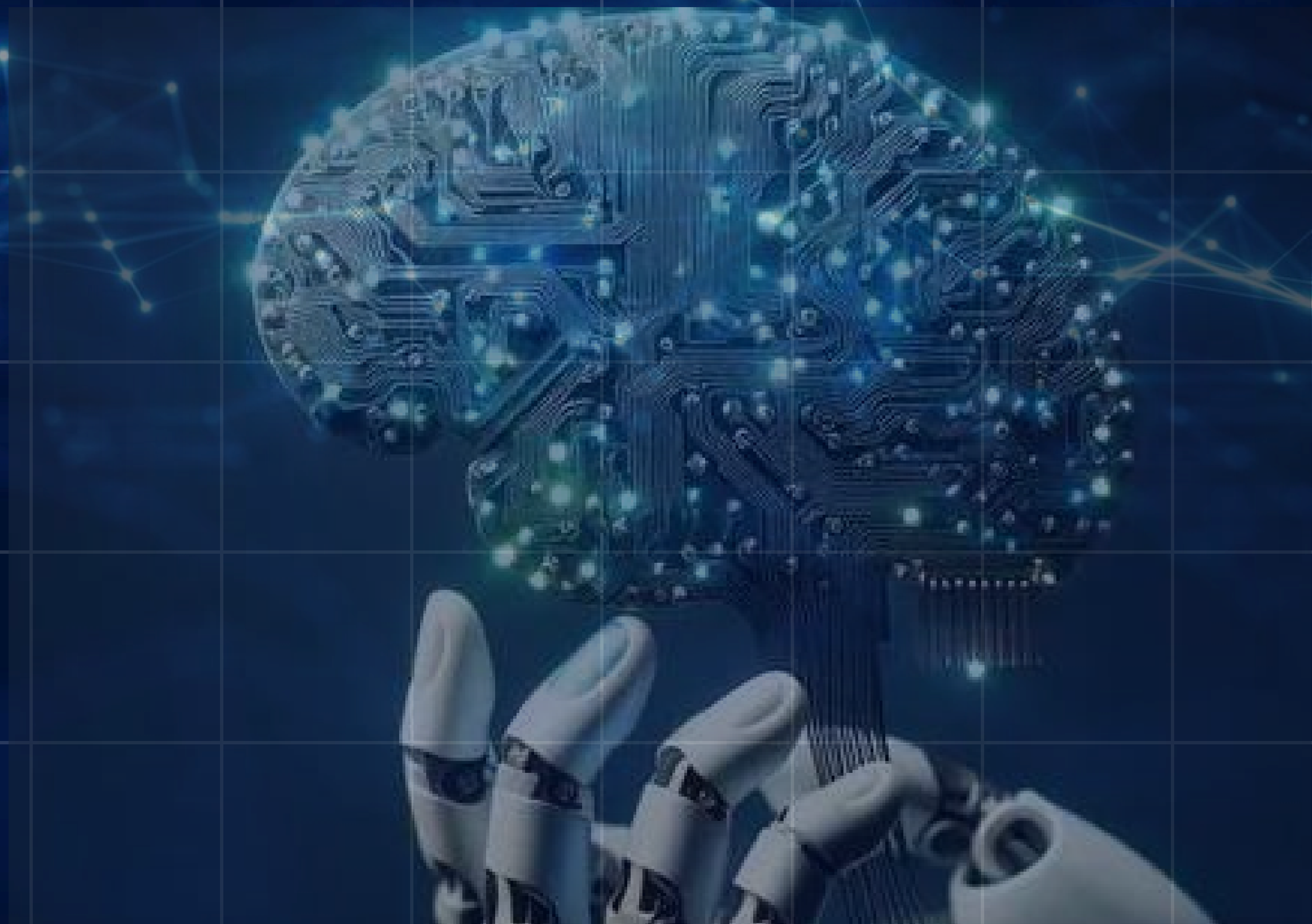
1 . Engineering is the application of science , mathematics, and improve products ,machines, structures ans sytems that solve real world problems .

2 . Engineers use creativity, technical knowledge, and innovation to make life safer, easier, and more efficient.

KEY POINTS

1 . Applies science and mathematics

2. Solves real-life problems Designs and develops technology



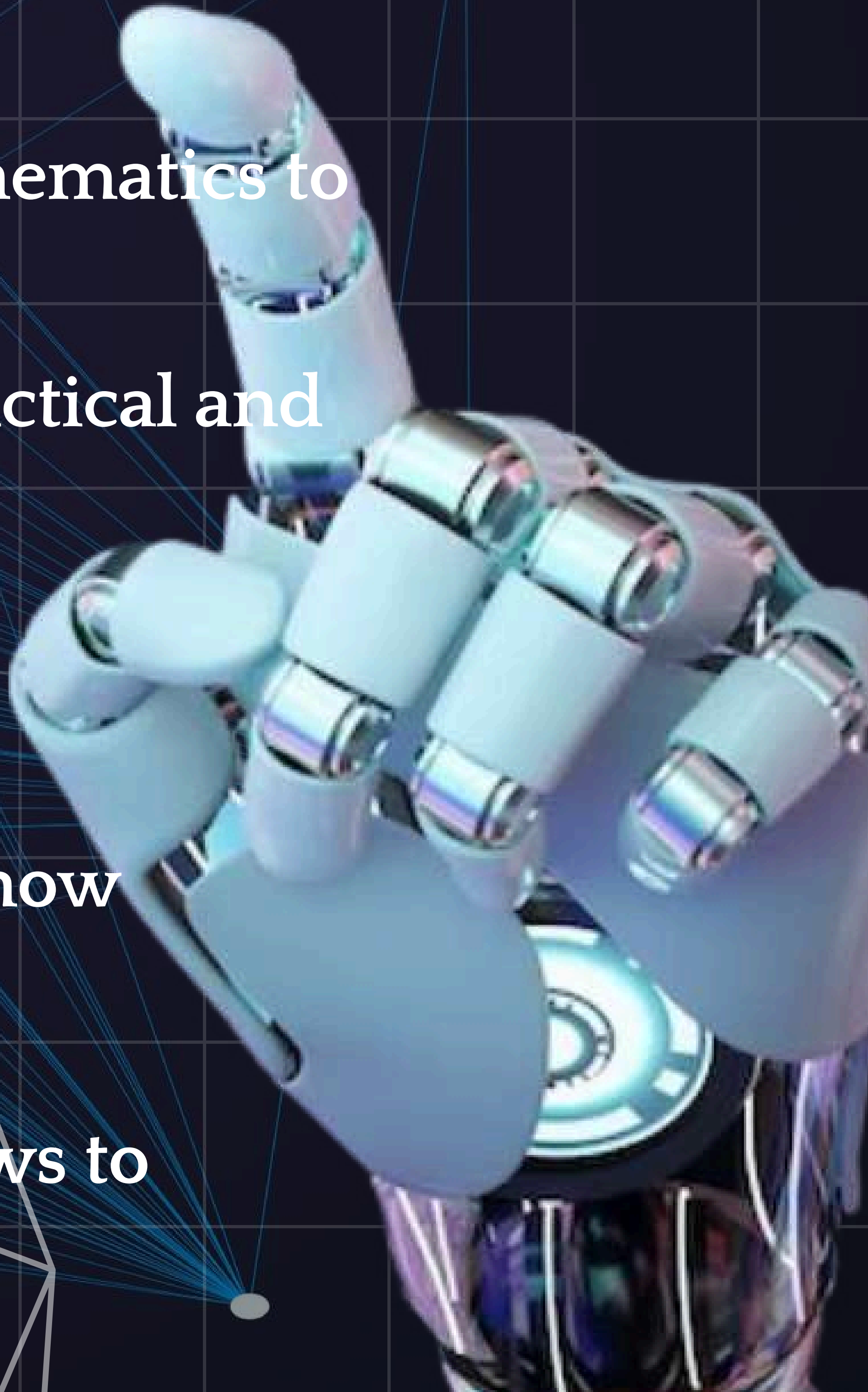
ENGINEERING VS SCIENCE

Engineering

- Engineering is the application of scientific knowledge and mathematics to design, build, and improve products, systems, and technologies
- Engineers focus on solving real-world problems by creating practical and efficient solutions that meet human needs.

Science

- Science is the study of the natural world. It aims to understand how things work by observing, experimenting, and discovering new knowledge.
- Scientists ask questions, test ideas, and develop theories and laws to explain natural phenomena.



ENGINEER VS SCIENIST

ENGINEER

- Engineer: Focuses on applying scientific principles to solve practical problems. They design, build, and optimize systems, structures, and devices.

- Engineer: Aims to develop practical, efficient, and cost-effective solutions to real-world problems.

SCIENIST

- Scientist: Primarily concerned with understanding natural phenomena. They seek to discover new knowledge through research, observation, and experimentation.

- Scientist: Aims to expand human knowledge and understanding of the universe, often without immediate consideration for practical applications.

NEEDS AND WANTS

NEEDS

- Automate industries using AI and robotics.
- Protect data through cybersecurity.
- Improve communication and connectivity.
- Develop software and mobile applications.
- Support healthcare, education, banking, and transportation.

WANTS

- Sustainable and energy-efficient solutions.
- Innovation in AI, cloud computing, and IoT.
- Better user experience in apps and websites.
- Secure and reliable digital systems.

VARIOUS DISCIPLINES OF ENGINEERING

- Computer Science and Engineering
- Mechanical Engineering
- Civil Engineering
- Electrical Engineering (EE)
- Electronics and Communication Engineering (ECE)
- Information Technology (IT)
- Chemical Engineering
- Aerospace Engineering
- Aeronautical Engineering

AI BASED ROBOTIC SURGERY

INTRODUCTUON

- Modern healthcare requires highly accurate and minimally invasive surgeries to improve patient safety and reduce recovery time. Traditional surgeries may involve larger incisions, longer hospital stays, and a higher risk of complications.

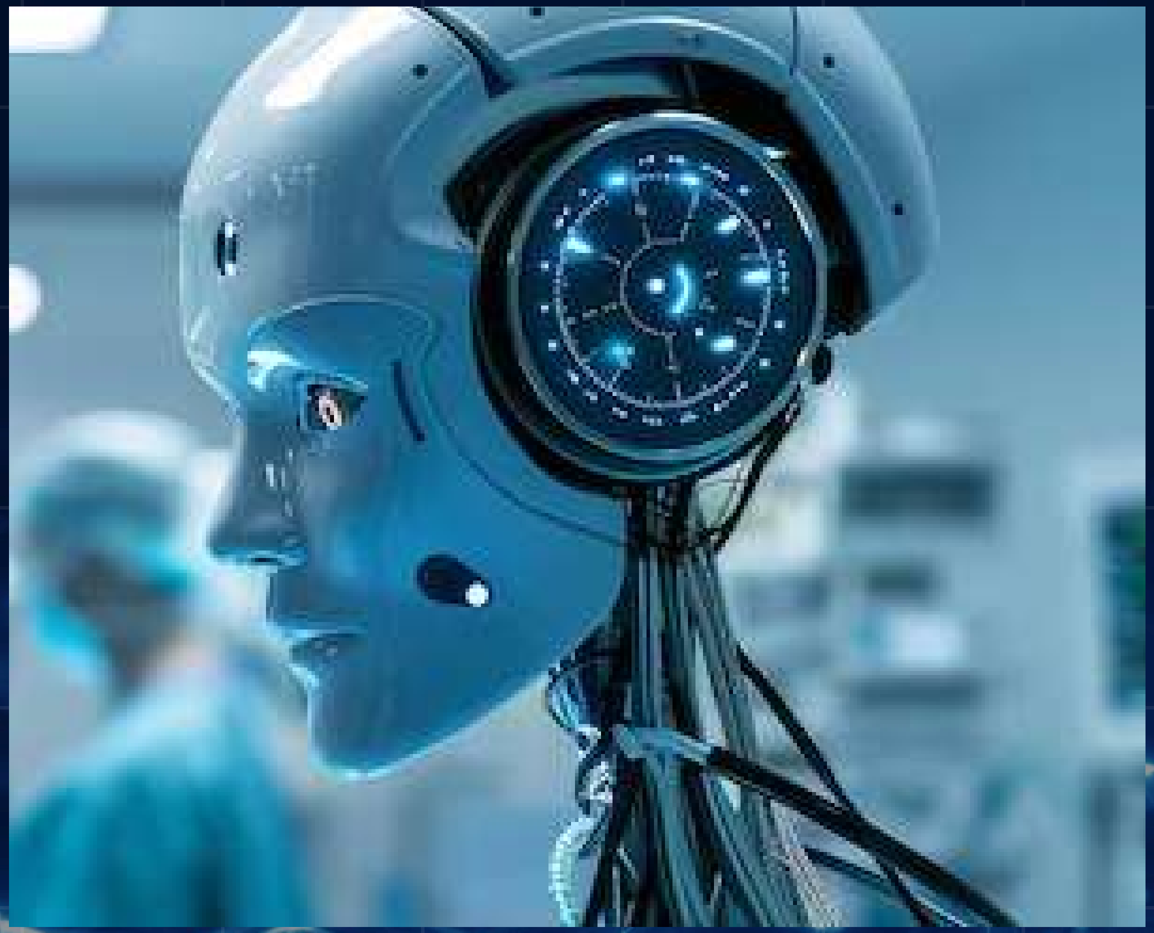
- An AI-Based Robotic Surgery System helps surgeons perform complex operations with greater precision using Artificial Intelligence (AI), robotics, computer vision, sensors, and real-time data processing. The surgeon controls the robotic arms through a computer console, while the system provides a clear 3D view and precise instrument movement.

COMPARISON TO COMPUTER SCIENCE ENGINEERING

- Computer Science Engineering plays a vital role in developing the intelligence behind the robotic surgery system by:
- Developing AI algorithms to assist surgeons during operations.
- Processing real-time images from high-definition cameras.
- Controlling robotic movements with precision.
- Ensuring secure storage of patient data through cloud technology.
- Creating software interfaces for surgeons to monitor and control the robot.

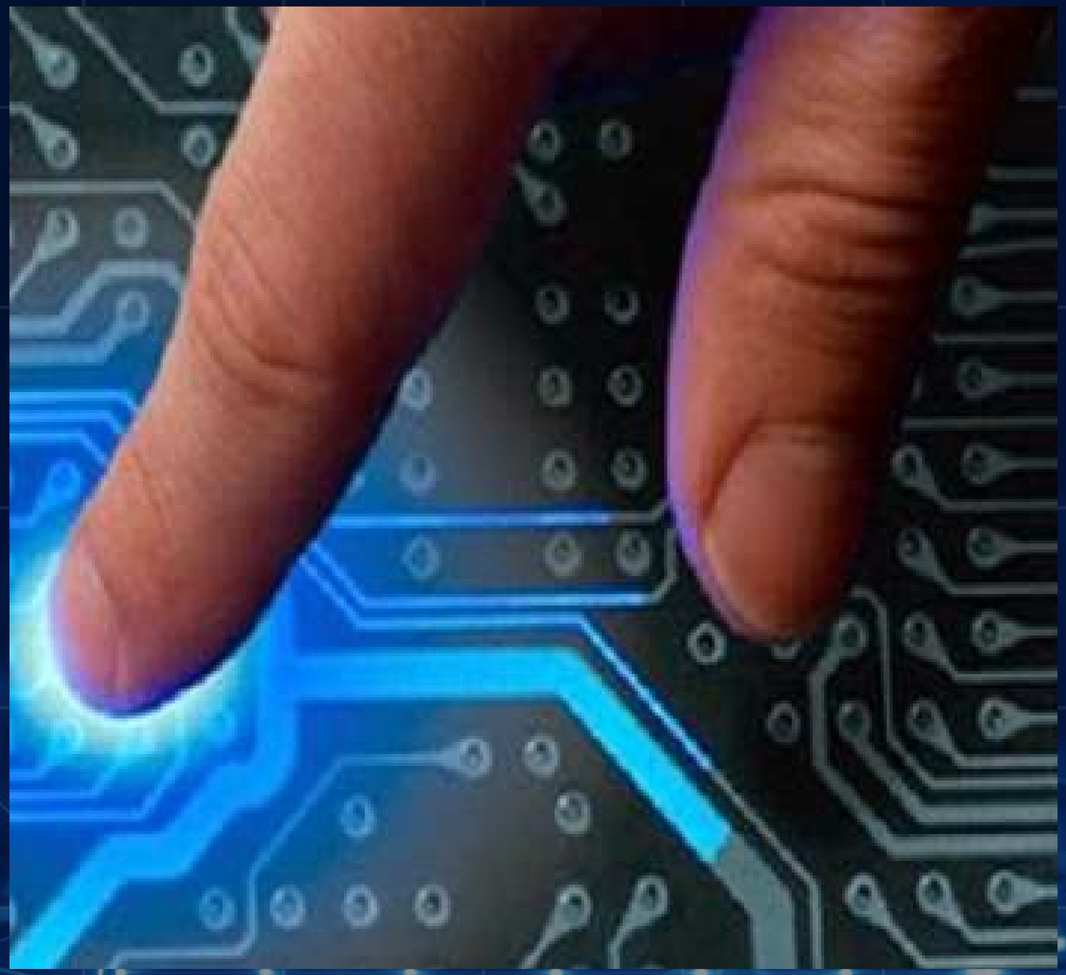
VARIOUS FIELD INVOLVED

ARTIFICIAL INTELLIGANCE



- Designs surgical instruments and robotic arms.
- Ensures the robot is safe for use in the human body.
- Works with doctors to improve surgical accuracy.

ELECTRICAL ENGINEERING



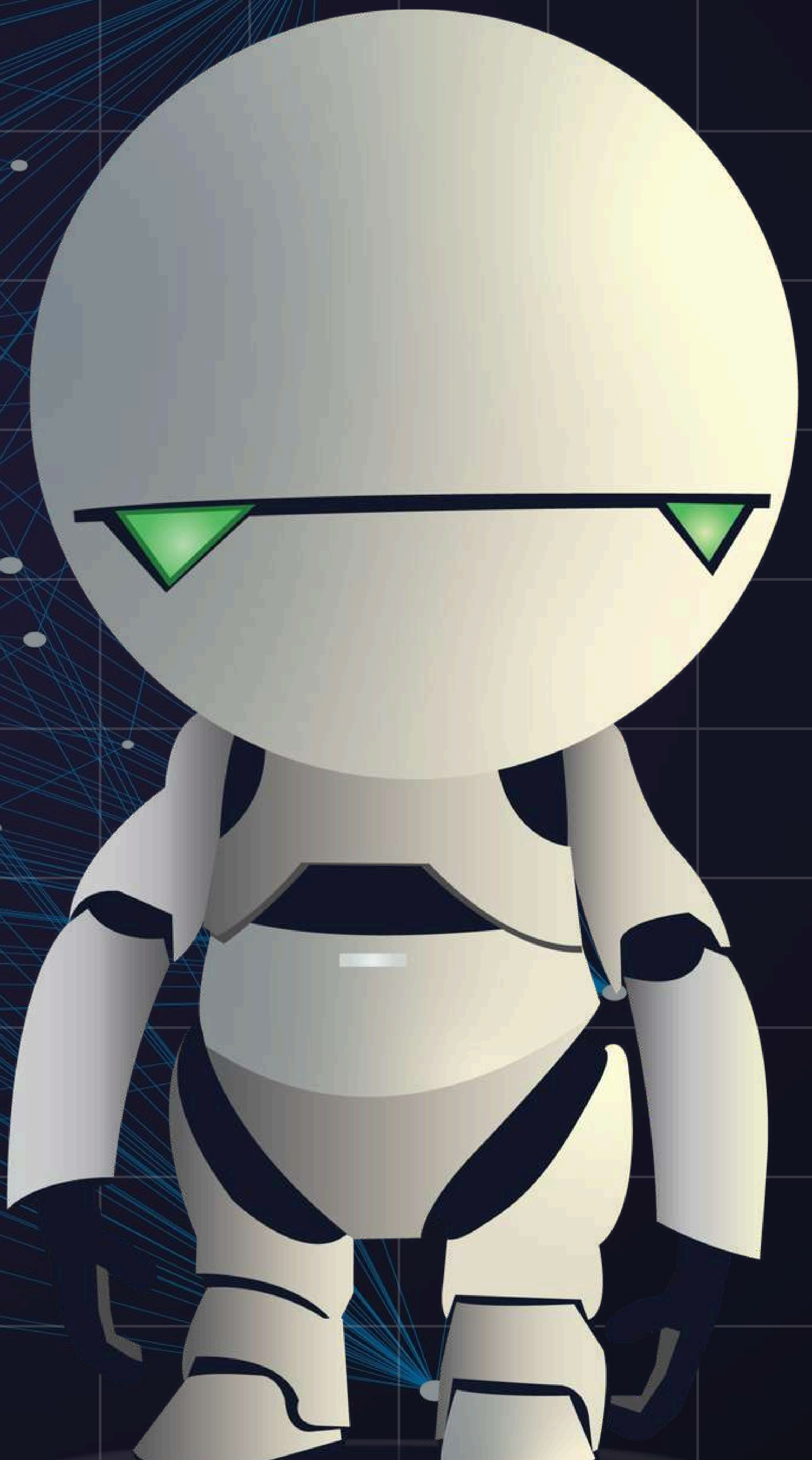
- upplies power to the robotic system.
- Designs sensors, motors, and control circuits.
- Ensains smooth movement of robotic arms.

BIO MEDICAL FIELD

- Analyzes medical images (CT, MRI, X-ray) to assist surgeons.
- Provides real-time guidance during surgery.
- Detects tissues, blood vessels, and tumors.

SKILLS REQUIRED FOT 21ST CENTURY

- CREATIVITY AND INOVATION
- PROBLEM ANALYSIS
- LIFELONG LEARNING
- FOLLOWING ETHICAL GUIDELINES
- COMMUNICATION SKILLS
- ENVIRONMENT AND SOCIAL
- RESPONSIBILITY



MISCONCEPTION OF CSE

Artificial Intelligence can completely replace Computer Science Engineers.

* Reality: AI assists engineers, but humans are needed to design algorithms, ensure ethical use, solve new problems, and improve AI systems.

Computer Science Engineers only build software.

- Reality: They also design embedded systems, IoT devices, cybersecurity solutions, cloud platforms, robotics software, and intelligent automation systems.

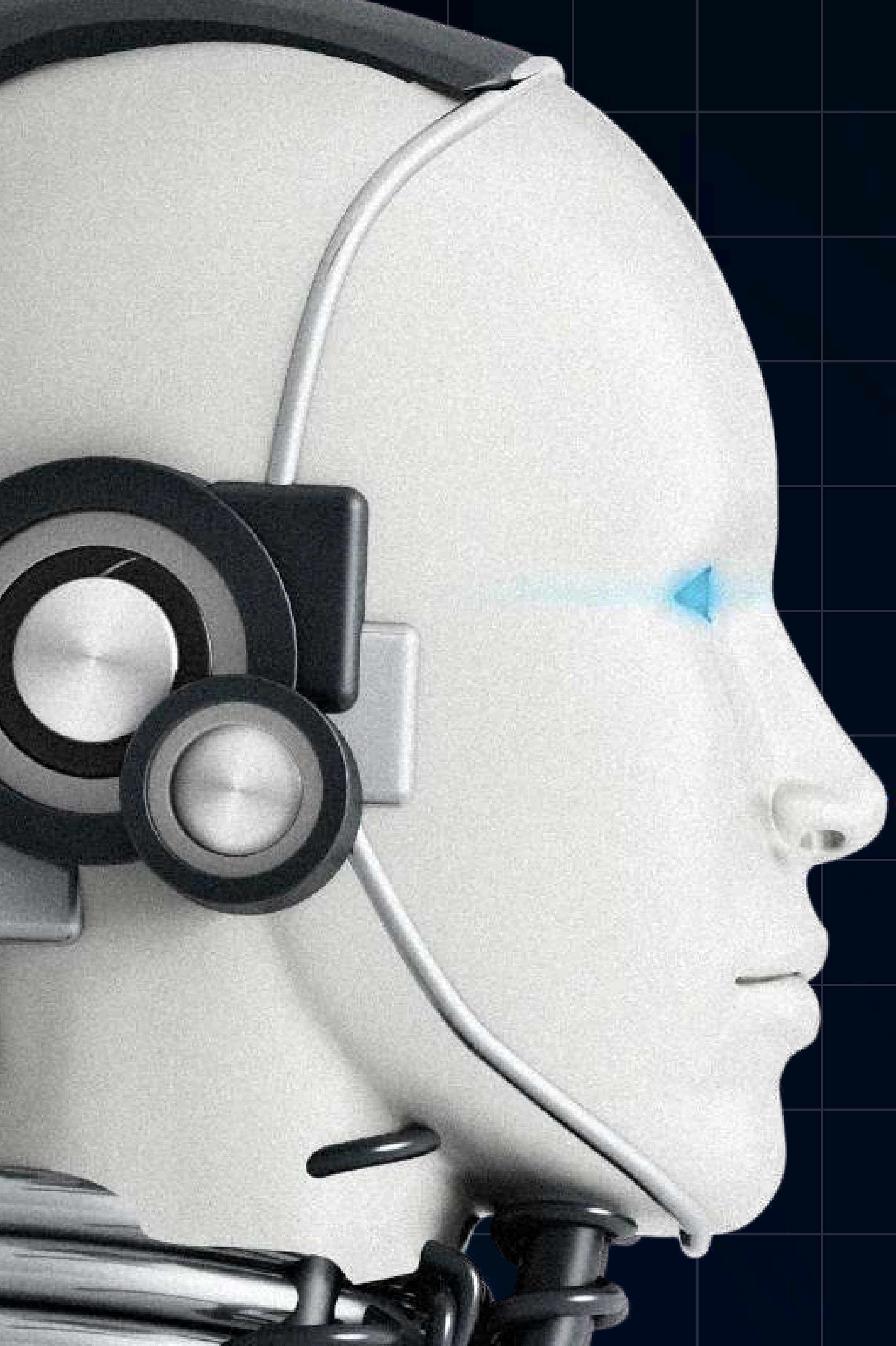
Computer Science Engineers only work for software companies.

- Reality: Today, almost every industry — including agriculture, healthcare, banking, aerospace, automotive, defense, manufacturing, and entertainment — needs Computer Science Engineers.

ABOUT CANVA

- Canva AI can be used to create presentations, videos, documents, websites, and posters. It also helps edit designs, improve image quality, and make designs look better using AI.

- For my presentation, I used an existing Canva template. I edited the text, changed the design, added my own images and content, and customized it to match my topic.



Thank You